Off The Grid: Efficient Trajectory Discovery with Adaptive Batch Sampling

DMDU Annual Meeting 2026

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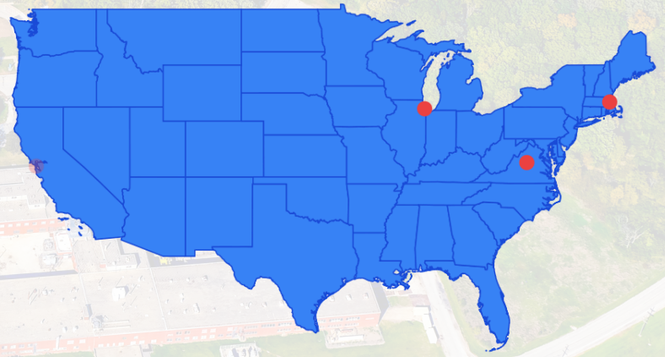


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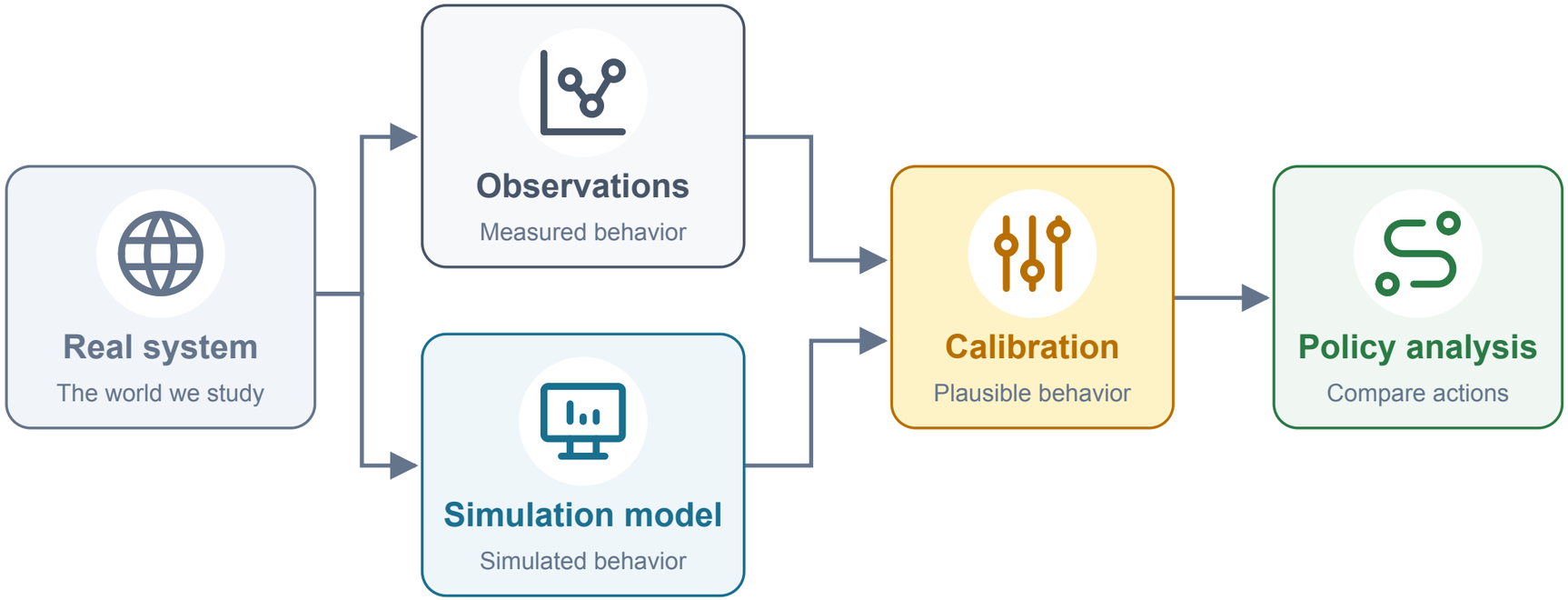




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Our Setting

Simulation-based inference supports policy analysis under uncertainty.

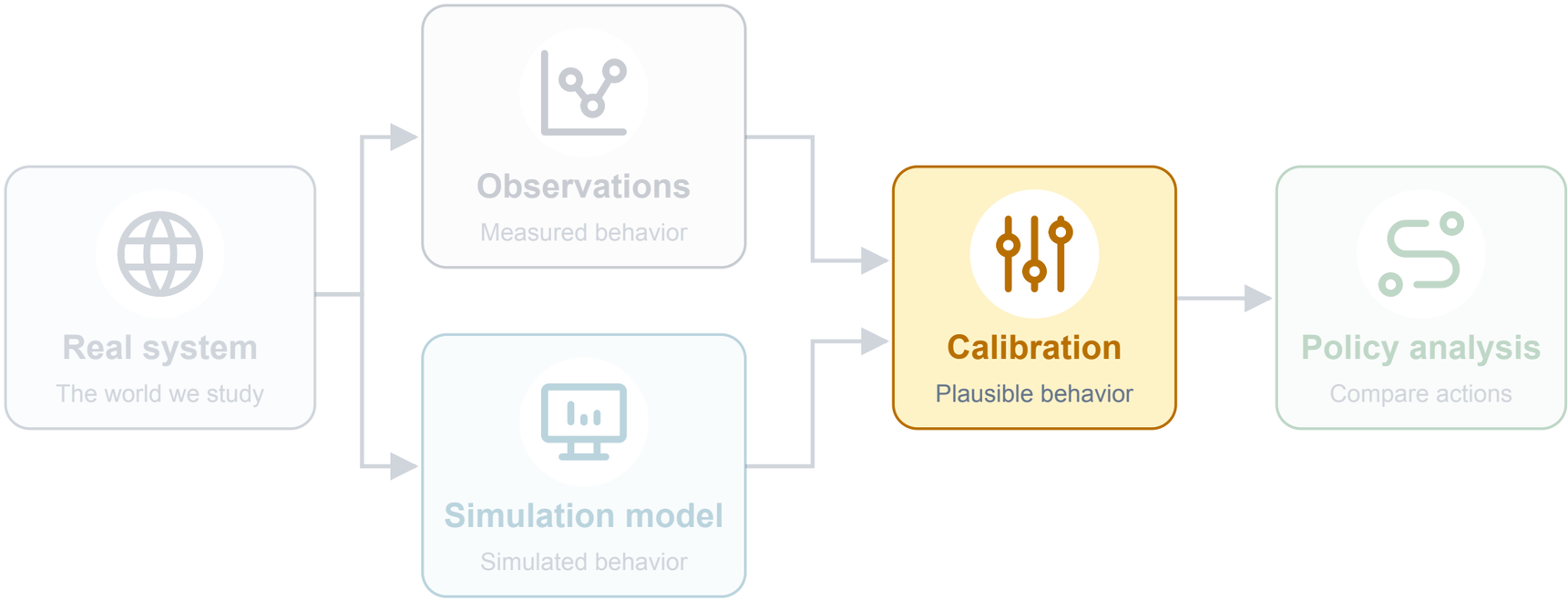


Calibration shapes the futures we carry into policy analysis.

Icon sources: world, chart-dots, desktop-analytics, adjustments, route — Tabler Icons, Paweł Kuna; MIT license. Adapted colors and size.

Calibration

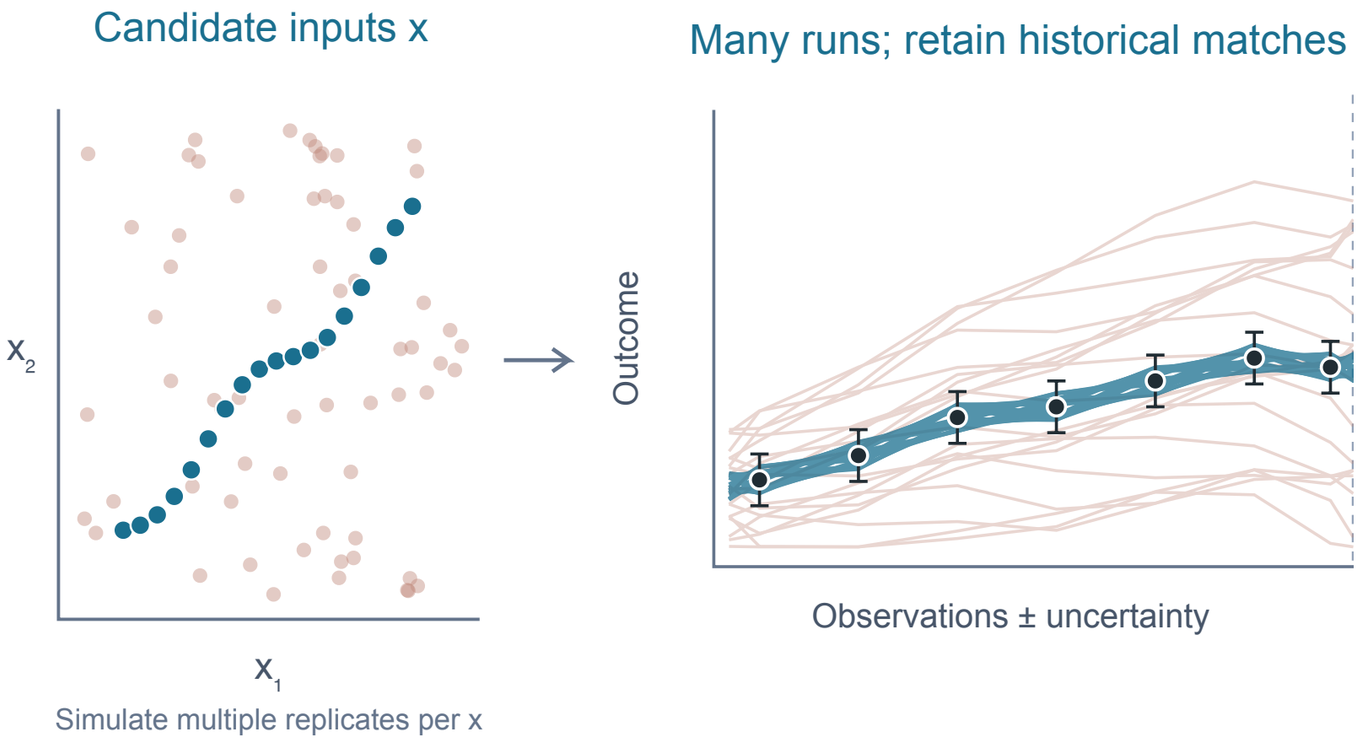
Our focus: constrain simulation behavior using observations.



Calibration shapes the futures we carry into policy analysis.

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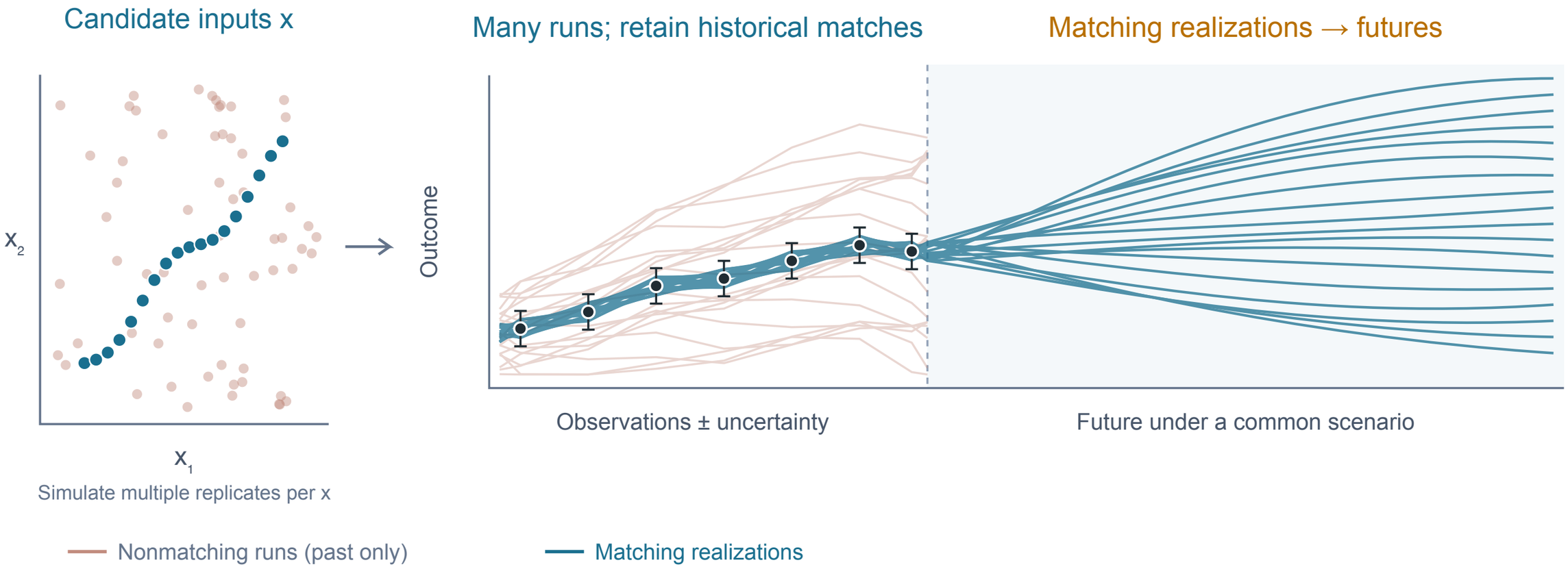
Calibration Shapes Decision Uncertainty



Conceptual illustration; input colors summarize matching (x,r) pairs

Search candidate inputs → Retain historical matches

Calibration Shapes Decision Uncertainty

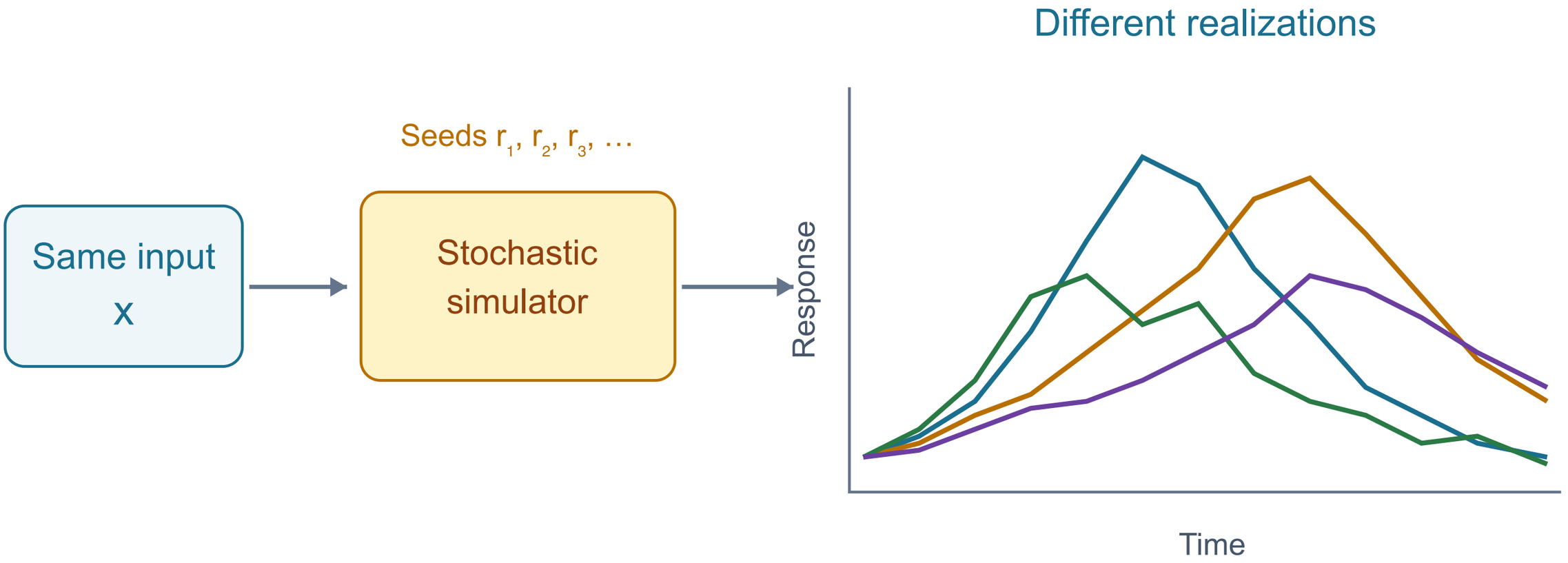


Conceptual illustration; input colors summarize matching (x,r) pairs

Calibrate to observations → Propagate uncertainty → Stress-test decisions

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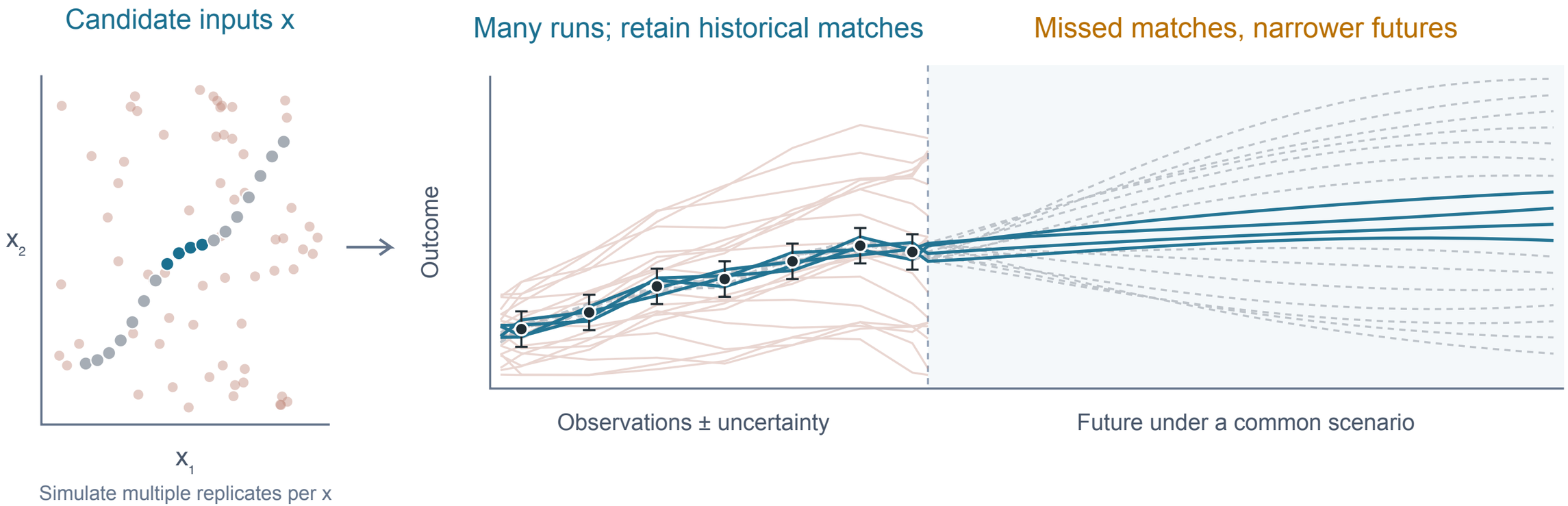
Stochastic Simulation



Conceptual illustration

$y(t) = f(x, r; t)$ — **Fixed input x, different random seeds r**

Missed Matches



Miss relevant (x, r) matches → Miss plausible futures

Finding Matches

Search over inputs and replicates

$$y(t) = f(x, r; t)$$

Keep pairs whose histories match the observations:

$$\square_\delta = \{(x, r) : d(f(x, r), y_{\text{obs}}) \leq \delta\}$$

Many compatible paths, rather than one best fit.

The computational challenge

- Many inputs x
- Many replicates r
- **Limited simulation budget**

Which (x, r) should we run next?

What Existing Methods Lack

Approximate Bayesian Computation

- Simulate, compare to data via summary statistics
- Accept/reject based on distance threshold
- **Potential gap:** compressed summaries can discard trajectory-level information

Mean-based Emulation + Calibration

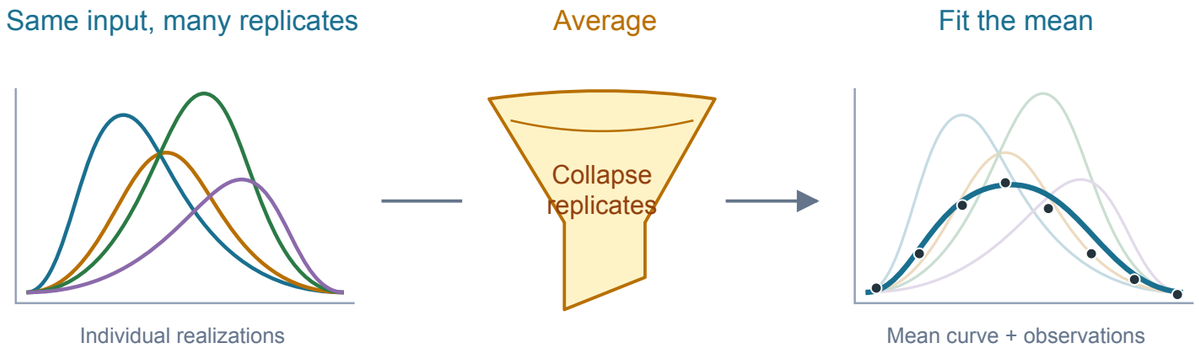
- Build a surrogate of $\mathbb{E}[f(x, r)]$
- Optimize over x
- **Gap:** averaging before emulation loses replicate-specific behavior

Mean-based methods **aggregate over randomness**. A good mean fit does not identify data-consistent realizations.

The consequence: the search targets the mean response, rather than explicitly discovering compatible (x, r) pairs.

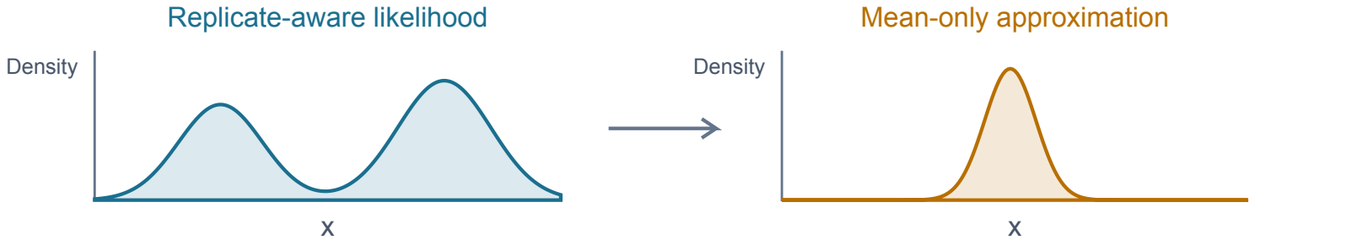
What Existing Methods Lack

Mean-based calibration **aggregates over randomness** before searching.



Which replicate matched? The average does not tell us.

The inferred distribution of x can change



Illustrative posterior shapes; not experimental results

Averaging can hide matches and **distort inference on x**.

Our Contribution

Trajectory-Oriented Optimization (TOO)

Replicate-aware surrogate

- Learn individual realizations
- Share information across replicates
- Predict untried (x, r) pairs

Adaptive grid Thompson sampling

- Explore promising regions
- Refine candidate locations
- Select simulations in batches

Replicate-Aware Surrogate

Common Random Number Gaussian Process (CRNGP)

A surrogate for individual realizations, indexed by input x and replicate r:

$$f(x,r) \sim \mathrm{GP}\Big(m(x,r),\ k\big((x,r),(x',r')\big)\Big)$$

Shared structure

$$k\big((x,r),(x',r')\big) = k_x(x,x')\,k_r(r,r')$$

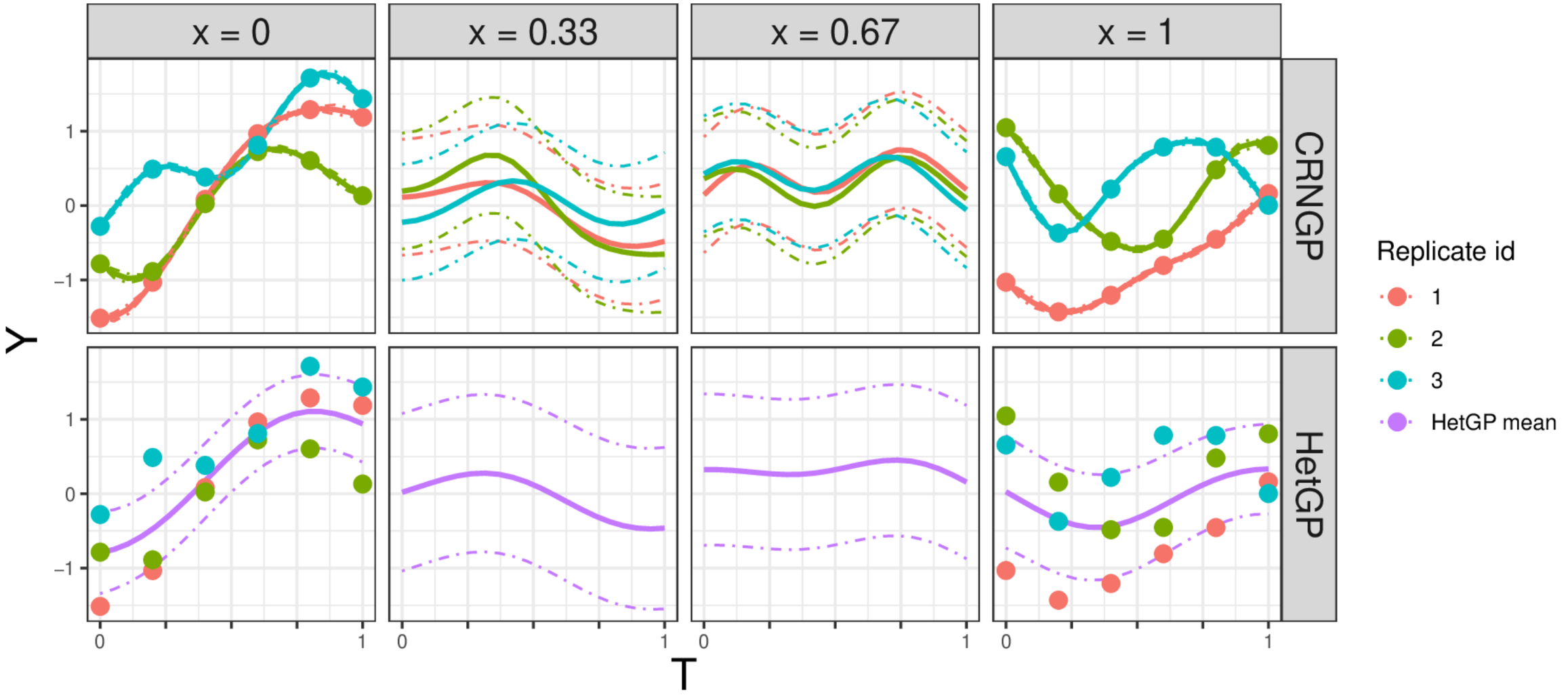
- Learn how responses vary with x
- **Borrow information across replicates**

Replicate-specific predictions

$$f(x,r) \mid \square \sim \square\left(\mu(x,r),\sigma^2(x,r)\right)$$

- Predict each realization's response
- **Quantify surrogate uncertainty** at untried (x,r)

CRNGP in Practice



Adaptive Grid Thompson Sampling

Choose the next batch of (x, r) pairs; refine where we search.

Sample

- Fit/update surrogate based on $f_n = \{f(x_1, r_1), \ldots, f(x_n, r_n)\}$.
- Draw plausible response from the CRNGP posterior at many (x_b, r_b):

$$\widetilde{f}(x_i, r_i) \sim p(f \mid \mathrm{f}_n)$$

Select & simulate

- Each draw proposes a promising pair from the grid $\square_n$.

$$(x_b, r_b) = \arg\min_{(x,r)\in\square_n} |\widetilde{f}(x, r) - y_{\text{obs}}|$$

- Run new simulation at (x_b, r_b)
- Update CRNGP surrogate

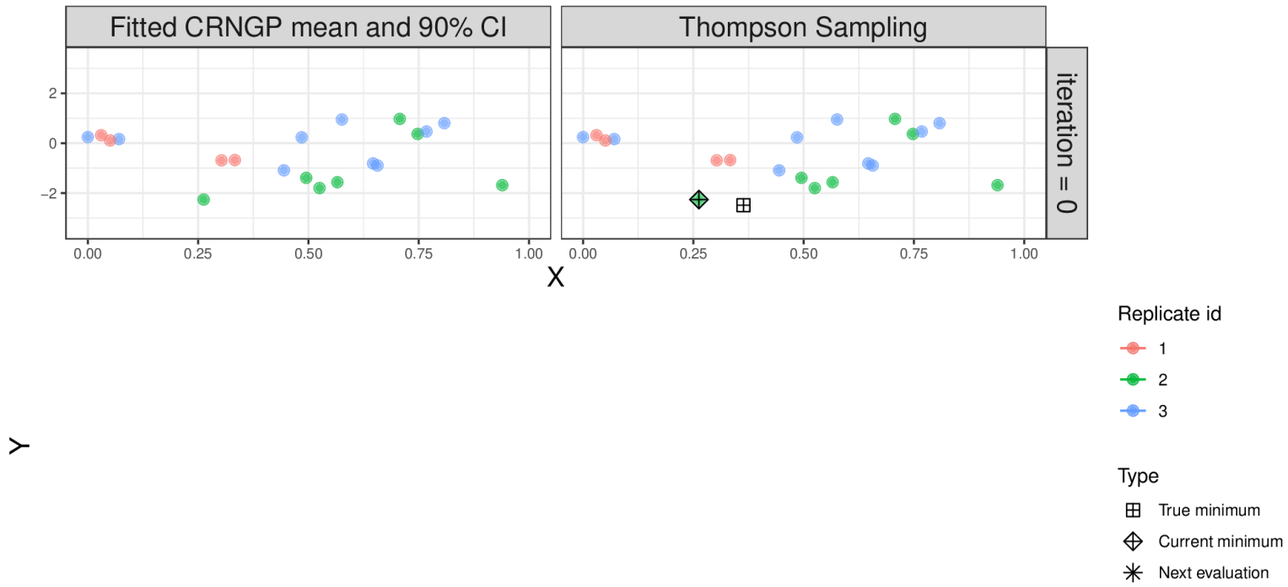
Adapt

- Refine the candidate grid:

$$\square_n \longrightarrow \square_{n+1}$$

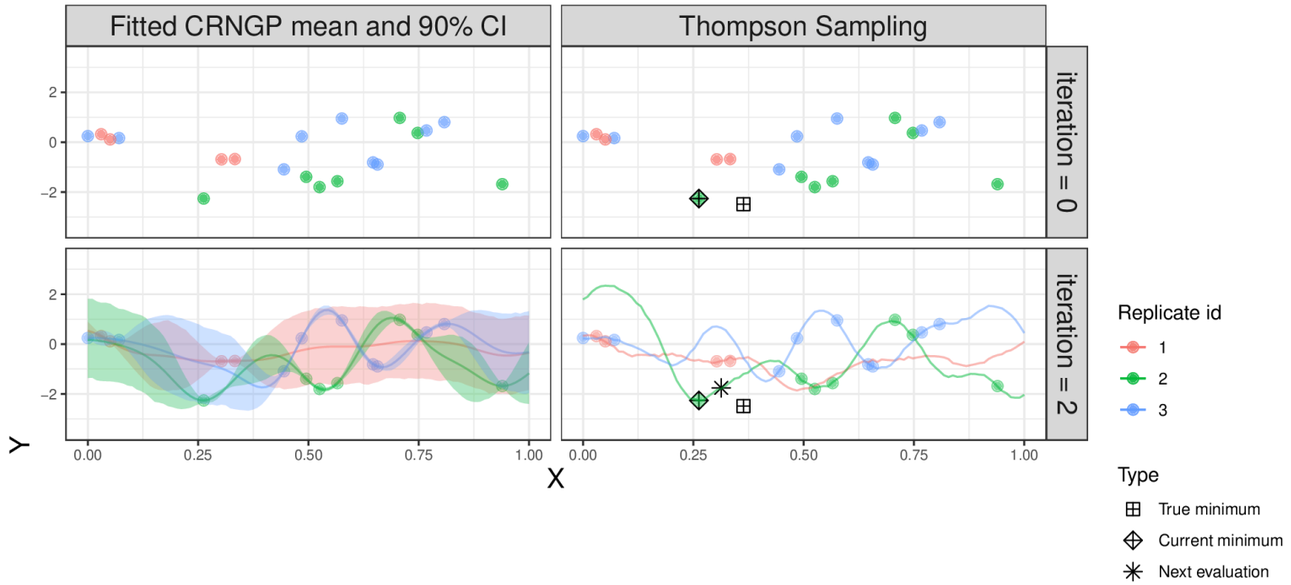
Concentrate and add candidates near promising regions.

Fixed Grid Thompson Sampling



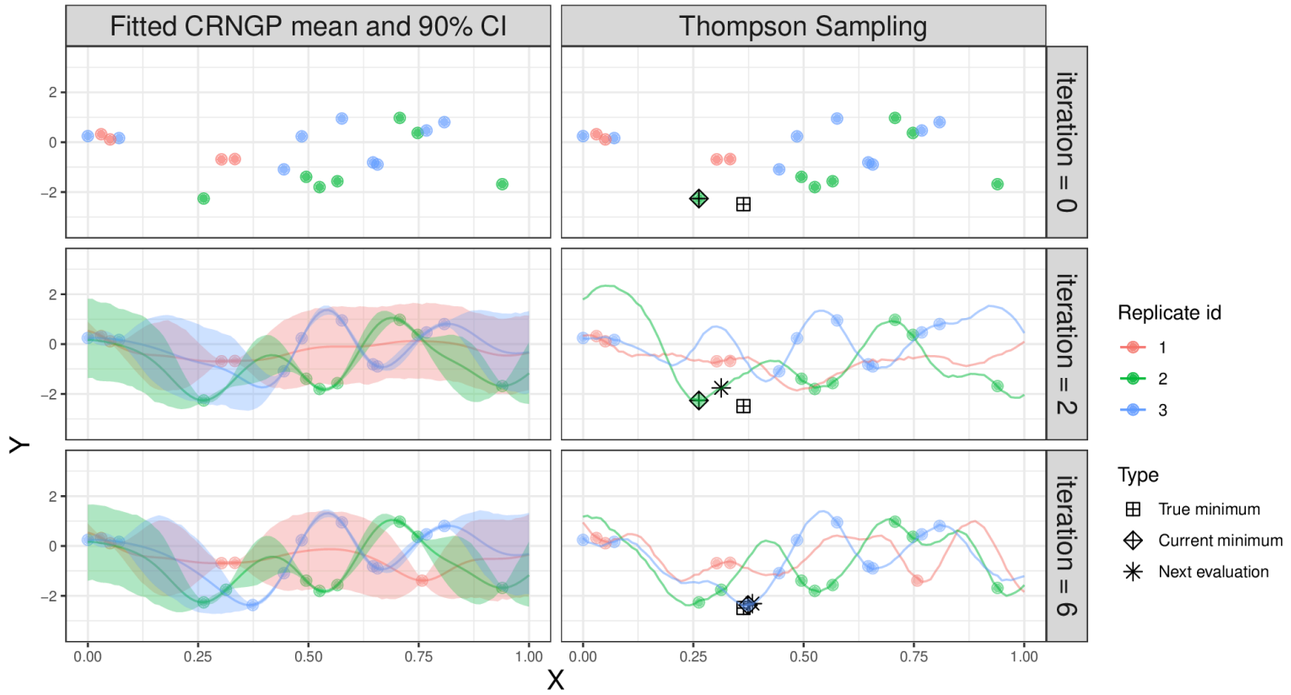
First 1 iteration rows shown. Thompson Sampling iteratively selects (x, r) candidates by sampling from the CRNGP posterior and picking the point with lowest predicted discrepancy.

Fixed Grid Thompson Sampling



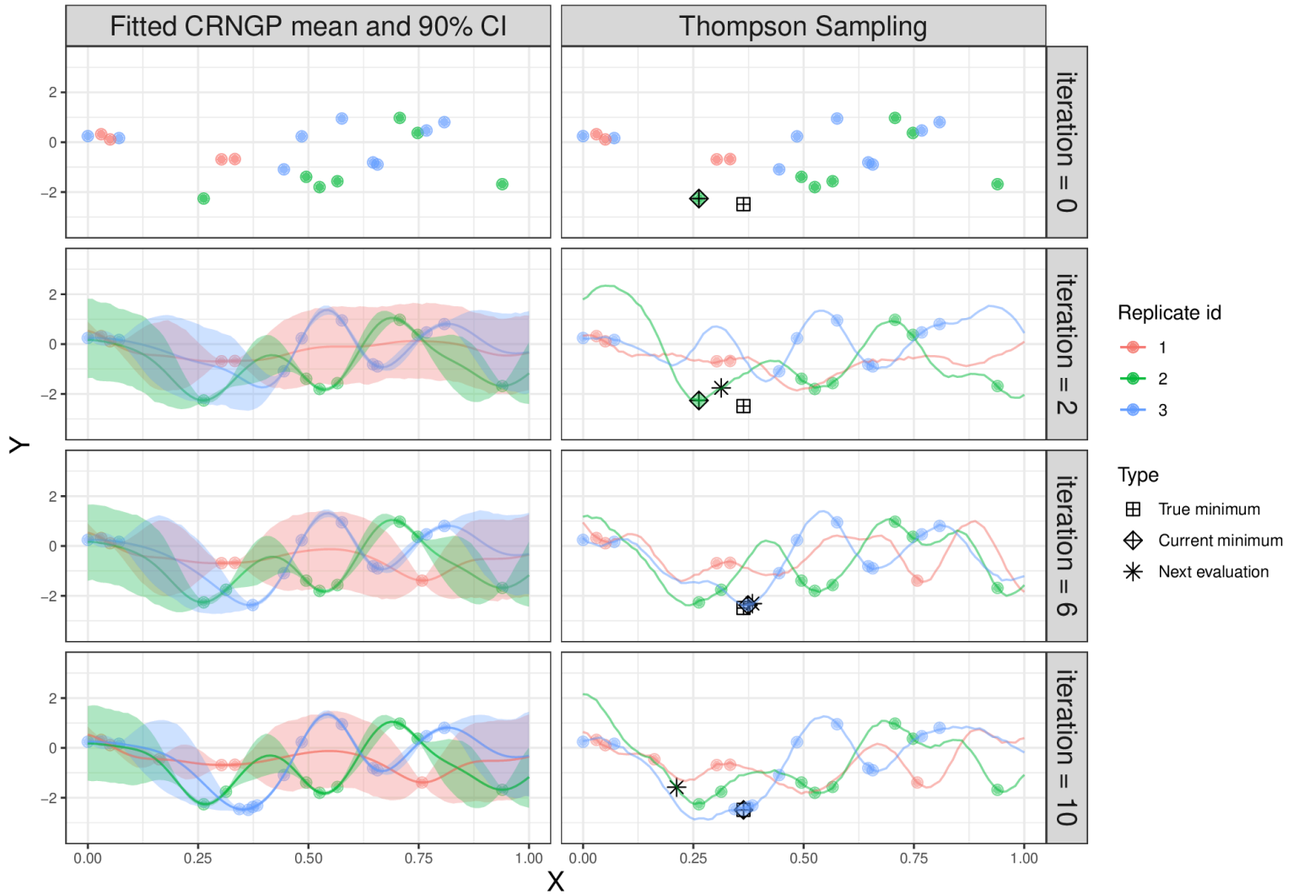
First 2 iteration rows shown. Thompson Sampling iteratively selects (x, r) candidates by sampling from the CRNGP posterior and picking the point with lowest predicted discrepancy.

Fixed Grid Thompson Sampling



First 3 iteration rows shown. Thompson Sampling iteratively selects (x, r) candidates by sampling from the CRNGP posterior and picking the point with lowest predicted discrepancy.

Fixed Grid Thompson Sampling



First 4 iteration rows shown. Thompson Sampling iteratively selects (x, r) candidates by sampling from the CRNGP posterior and picking the point with lowest predicted discrepancy.

Adaptive Grid Thompson Sampling

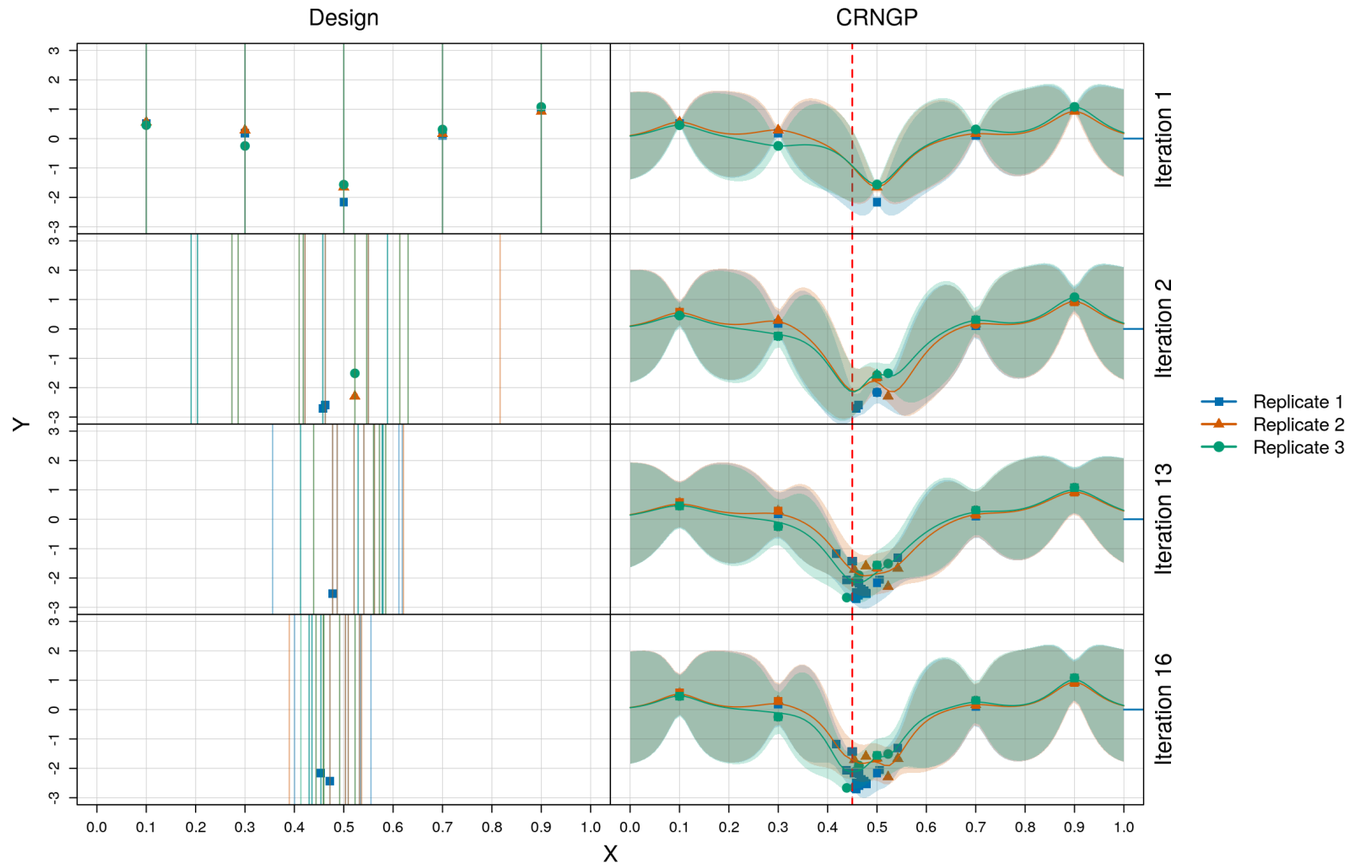
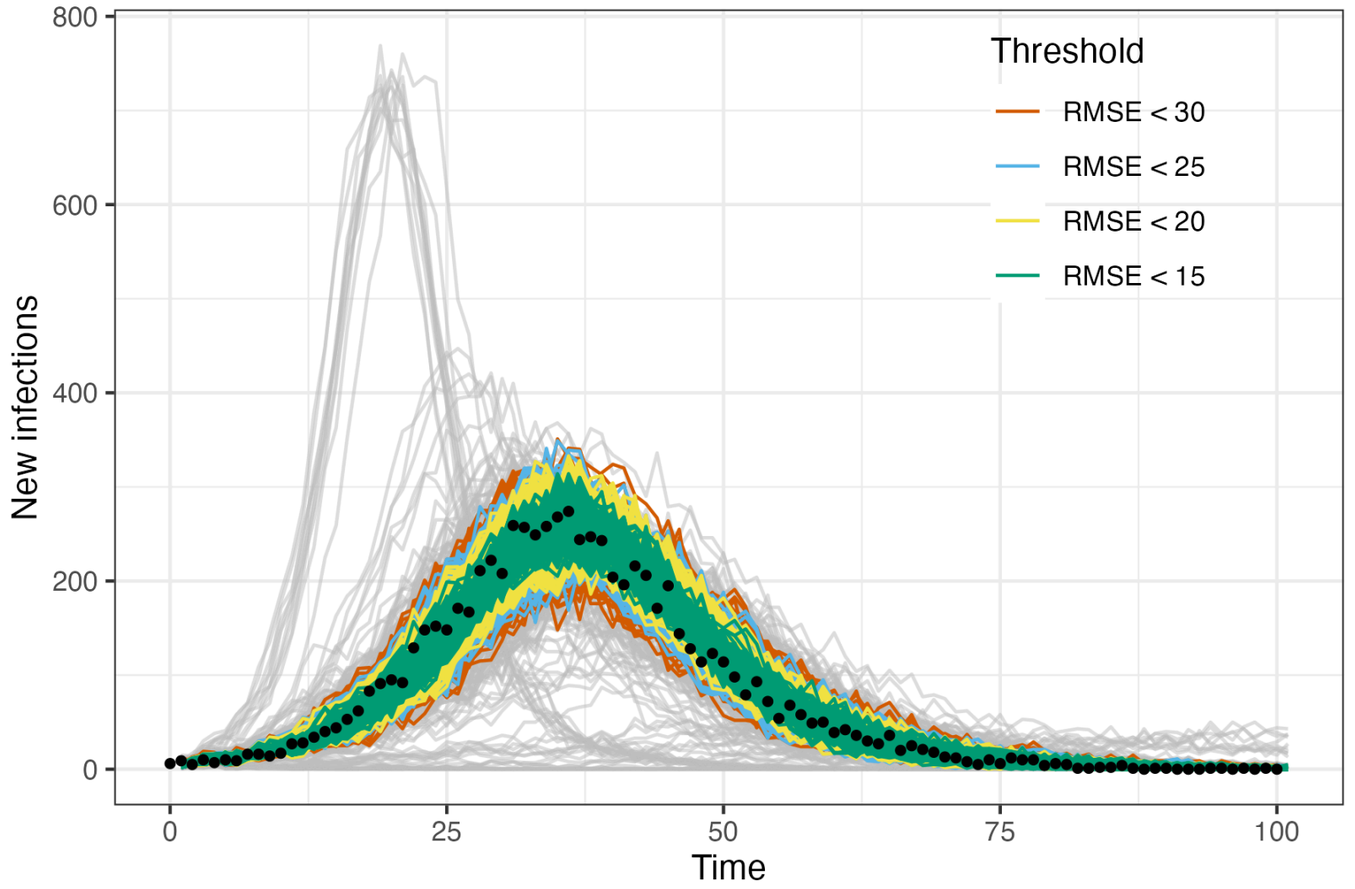


Illustration of adaptive grid TS with a CRNGP surrogate to find (X, r) tuple having minimal Y values.

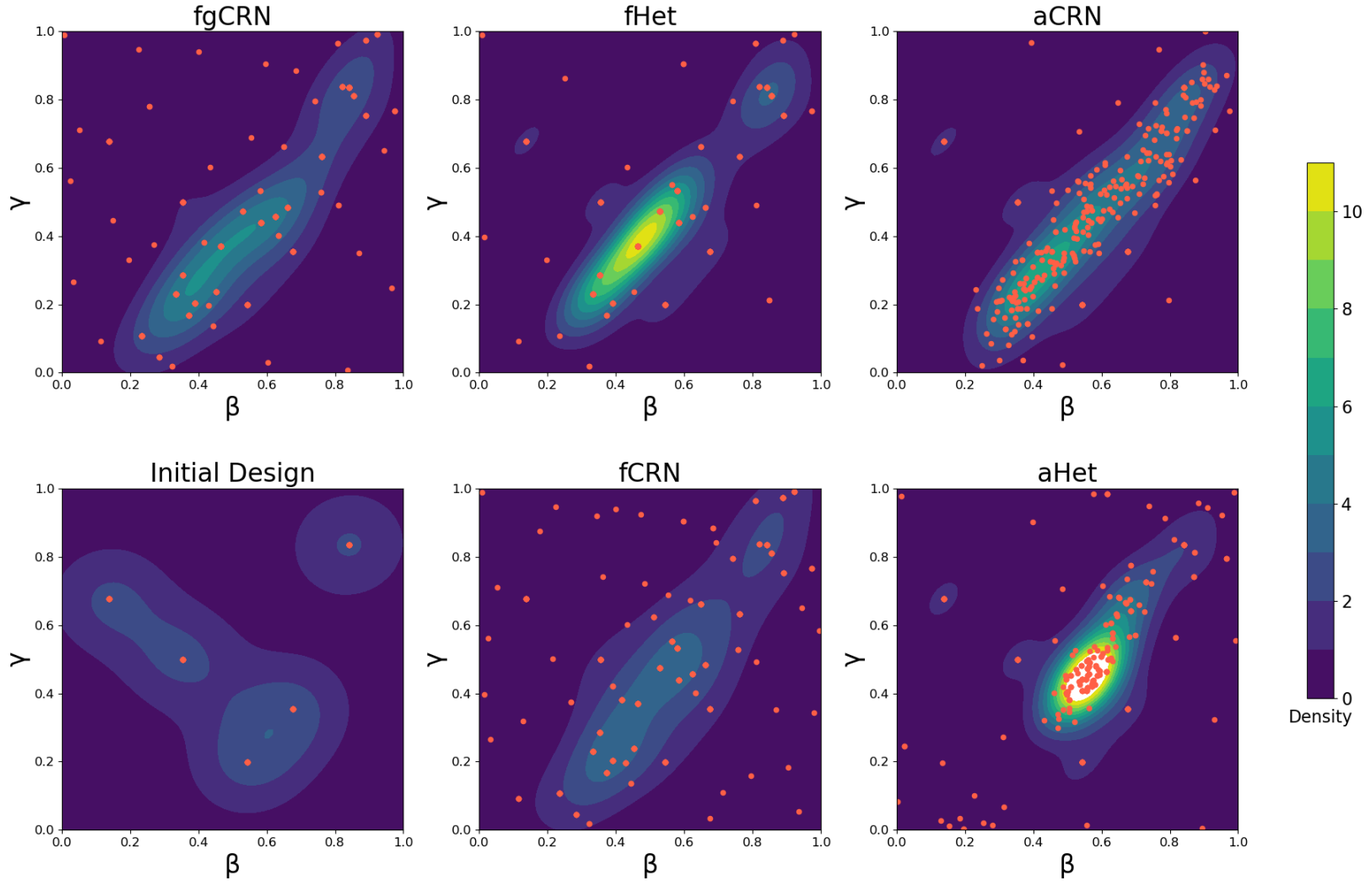
Result - Matching Trajectories

Stochastic SIR: multiple realizations reproduce the observed infection curve.



Result - Exploring Inputs

aCRN explores an extended region of infection and recovery rates.



Summary

- **Calibration matters downstream.** Missed historical matches can mean missed plausible futures.
- **Search for realizations.** TOO targets compatible (x, r) pairs, rather than only a mean response.
- **Adaptive optimization approach.** aCRN combines a replicate-aware surrogate with adaptive grid Thompson sampling.
- **DMDU relevance:** carry diverse, data-consistent behaviors forward to stress-test policies across plausible futures.

Staying on Track: Efficient Trajectory Discovery with Adaptive Batch Sampling

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